ECEN-629: Homework 4

Due on November 8, 2025 at 11:59 pm

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Problem 1

Consider the problem

$$\begin{aligned} \text{minimize:} \quad & x_1^2 + x_2^2 \\ \text{subject to:} \quad & (x_1 - 1)^2 + (x_2 - 2)^2 \leq 2 \\ & 2x_1 + x_2 \geq 3 \\ & x_1 \geq 0,\, x_2 \geq 0. \end{aligned}$$

(a) Derive the KKT conditions. Find a solution for these conditions. Is this the optimal solution?

(b) Does the strong duality hold?

(c) Compare the solution to the one you obtained last time using geometry. Are they the same?

(d) Consider the problem

$$\begin{aligned} \text{minimize:} \quad & x_1^2 + x_2^2 \\ \text{subject to:} \quad & (x_1 - 1)^2 + (x_2 - 2)^2 \leq 2 + \mu_1 \\ & 2x_1 + x_2 \geq 3 + \mu_2 \\ & x_1 \geq 0,\, x_2 \geq 0. \end{aligned}$$

The optimal value is a function of μ_1 and μ_2, i.e., $p^*(\mu_1, \mu_2)$. Compute

$$\left.\frac{\partial p^*(\mu_1, \mu_2)}{\partial \mu_1}\right|_{(0,0)}$$

$$\left.\frac{\partial p^*(\mu_1, \mu_2)}{\partial \mu_2}\right|_{(0,0)}$$

Solution

Part (a): The Lagrangian:

$$L(x, \lambda) = x_1^2 + x_2^2 + \lambda_1 \left((x_1 - 1)^2 + (x_2 - 2)^2 - 2\right) + \lambda_2(3 - 2x_1 - x_2) + \lambda_3(-x_1) + \lambda_4(-x_2)$$

and our conditions:

$$\begin{cases} \lambda_1 \left((x_1 - 1)^2 + (x_2 - 2)^2 - 2\right) = 0 \\ \lambda_2(3 - 2x_1 - x_2) = 0 \\ \lambda_3(-x_1) = 0 \\ \lambda_4(-x_2) = 0 \\ 2x_1 + 2\lambda_1(x_1 - 1) - 2\lambda_2 - \lambda_3 = 0 \\ 2x_2 + 2\lambda_1(x_2 - 2) - \lambda_2 - \lambda_4 = 0 \\ \lambda \succeq 0 \end{cases}$$

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Part (b): The strong duality holds since Slater's condition is satisfied.

Part (c): This form has the same solution as the solution obtained by geometric analysis $x^\star = (1.2, 0.6)^T$.

Part (d): By local sensitivity, we have

$$\lambda_i^* = -\frac{\partial p^*(\mu_1, \mu_2)}{\partial \mu_i}\bigg|_{(0,0)}$$

Therefore

$$\frac{\partial p^*(\mu_1, \mu_2)}{\partial \mu_1}\bigg|_{(0,0)} = -\lambda_1^* = 0 \quad \text{and} \quad \frac{\partial p^*(\mu_1, \mu_2)}{\partial \mu_2}\bigg|_{(0,0)} = -\lambda_2^*$$

Problem 2

Consider the following optimization problem:

$$\begin{aligned} \text{minimize:} \quad & \sum_{i=1}^{n} \alpha_i \max\left\{\ln \frac{\sigma_i^2}{x_i}, 0\right\} \\ \text{subject to:} \quad & \sum_{i=1}^{n} \beta_i x_i \leq D, \end{aligned}$$

where $x = (x_1, x_2, \ldots, x_n) \succ 0$ are the decision variables. Assume $\sigma^2 \succ 0$, $\alpha \succeq 0$, and $\beta \succ 0$.

(a) Argue that this problem is convex;

(b) Write down the KKT conditions;

(c) Provide an explicit solution for this problem (in terms of certain intermediate variable), by providing a solution to the KKT conditions;

(d) Show that equivalently, the inequality constraint can be replaced by an equality constraint for any $0 < D \leq \sum_{i=1}^{n} \beta_i \sigma_i^2$;

(e) Let $n = 3$, $(\sigma_1^2, \sigma_2^2, \sigma_3^2) = (1, 4, 25)$, $\alpha = (1, 1, 2)$, and $\beta = (2, 1, 1)$. Consider the optimal value p^* as a function of the parameter D. Find this function in a closed form.

(f) Design an algorithm to compute the optimal solution for any given $\sigma^2 \succ 0$, $\alpha \succeq 0$, and $\beta \succ 0$, with a computation complexity at most $O(n)$.

Solution

Part (a): This problem is convex as we can rewrite the max in the objective function as a constraint instead, which results in $\ln \frac{\sigma_i^2}{x_i}$ with the condition that $x_i \leq \sigma_i^2$.

Part (b): The Lagrangian:

$$L(x, \lambda) = \sum_{i=1}^{n} \alpha_i \ln \frac{\sigma_i^2}{x_i} + \lambda_1 \left(\sum_{i=1}^{n} \beta_i x_i - D \right) + \lambda_2 (x_i - \sigma_i^2)$$

and our KKT conditions:

$$\begin{cases} \lambda_1 \left(\sum_{i=1}^{n} \beta_i x_i - D \right) = 0 \\ \lambda_2 (x_i - \sigma_i^2) = 0 \\ \lambda \succeq 0 \\ \frac{-\alpha}{x_i} + \lambda_1 \beta_i + \lambda_2 x_i = 0 \end{cases}$$

Part (c): $x_i^* = -\alpha_i / \beta_i$.

Part (d): Since we impose the condition $x_i \leq \sigma_i^2$, then we can rewrite the original constraint condition as $0 < D \leq \sum_{i=1}^{n} \beta_i \sigma_i^2$.

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Part (e): $n = 3$, $(\sigma_1^2, \sigma_2^2, \sigma_3^2) = (1, 4, 25)$, $\alpha = (1, 1, 2)$, and $\beta = (2, 1, 1)$.

$$f_0(x) = \max\left\{\ln\frac{1}{x_1}, 0\right\} + \max\left\{\ln\frac{4}{x_2}, 0\right\} + 2\max\left\{\ln\frac{25}{x_3}, 0\right\}$$
$$f_1(x) = 2x_1 + x_2 + x_3 \leq D$$

Part (f): We want the largest x_i values that lie on the plane $2x_1 + x_2 + x_3 = D$.

Problem 3

Strong duality holds for the problem

$$\begin{aligned} \text{minimize:} &\quad -4x_1^2 + x_2^2 + x_3^2 + 2(x_1 + x_2 + x_3) \\ \text{subject to:} &\quad x_1^2 - 2x_2^2 + x_3^2 \leq 1, \end{aligned}$$

although the problem is not convex (see Appendix B.1). Derive the KKT conditions. Find all solutions for the KKT conditions. Which one is the optimal value (assuming primal and dual optimal values both attained)? You may need to use a program (Matlab, Mathematica, or Python package) to plot and solve a non-linear equation; approximate solution is acceptable.

Solution

The Lagrangian:

$$L(x, \lambda) = -4x_1^2 + x_2^2 + x_3^2 + 2(x_1 + x_2 + x_3) + \lambda(x_1^2 - 2x_2^2 + x_3^2 - 1)$$

KKT conditions:

(a) Primal constraints: $x_1^2 - 2x_2^2 + x_3^2 - 1 \leq 0$

(b) Dual constraints: $\lambda \succeq 0$

(c) Complementary slackness: $\lambda(x_1^2 - 2x_2^2 + x_3^2 - 1) = 0$

(d) Vanishing gradient:

$$\nabla f_0(x) + \lambda \nabla f_1(x) = 0 \quad \Longrightarrow \quad \begin{cases} -8x_1 + 2\lambda x_1 = 0 \\ 2x_2 + 2 - \lambda 4x_2 = 0 \\ 2x_3 + 2 + \lambda 2x_3 = 0 \end{cases}$$

Solving these equations, we get

$$x_1 = \frac{1}{4 - \lambda}, \quad x_2 = \frac{1}{2\lambda - 1}, \quad x_3 = \frac{-1}{\lambda + 1}.$$

Plugging these into the complementary slackness condition, we get

$$\lambda \left(\frac{1}{(4 - \lambda)^2} - 2\left(\frac{1}{2\lambda - 1}\right)^2 + \left(\frac{-1}{\lambda + 1}\right)^2 - 1 \right) = 0.$$

The solution (evaluated using WolframAlpha) is $\lambda \approx 3.0085$ and $\lambda \approx 5.00155$, which gives

$$x^\star \approx (1.00857, 0.199322, -0.24947)^T \quad \text{and} \quad x^\star \approx (-0.998452, 0.111073, -0.166624)^T.$$

Problem 4

Download the CVX package (with either the Python or Matlab interface), read the user manual, and then answer the following questions. The following constraints will cause an error in CVX as they will not be recognized as valid constraints in a convex optimization problem. Convert them to valid forms, and explain your answer. You may need to read part of the user guide of CVX.

(a) `norm([x + 2*y, x - y])==0;`

(b) `norm([max(x,1), max(y,2)]) <= 3*x + y;`

(c) `x^3 + y^3 <= 1; x >= 0; y >= 0;`

(d) `x*y >= 1; x >= 0; y >= 0.`

Solution

Part (a): This yields the following error: `Invalid constraint: convex == constant`. We could instead require both components of the vector to equal zero: `x + 2*y == 0; x - y == 0;`

Part (b): This yields the following error: `Cannot perform the operation norm( convex, 2 )`. We can replace the max functions with inequalities.

Part (c): This yields the following error: `Illegal operation: real affine .^3`. To fix this, we can use the `POW_POS` function: `POW_POS(x, 3) + POW_POS(y, 3) <= 1;`

Part (d): This yields the following error: `Invalid quadratic form(s): not a square.` We can fix this by using `geo_mean([x, y], 1) >= 1;` instead.

Problem 5

The gap function $\phi_r(x)$ measures the gap between the largest r component and smallest r component of the vector $x \in \mathbb{R}^n$, where $1 \leq r \leq n$. Consider the optimization problem

$$\begin{aligned} \text{minimize:} \quad & \|Ax - b\| \\ \text{subject to:} \quad & \phi_r(x) \leq \alpha. \end{aligned}$$

(a) Find an equivalent LP formulation of the problem when the norm is ℓ_1.

(b) With the same ℓ_1 norm, set

$$A = \begin{bmatrix} 1 & 2 & 3 & 6 & 9 \\ 2 & 3 & 7 & 2 & 4 \\ 4 & 2 & 2 & 5 & 1 \end{bmatrix}$$

and $b = [8, -20, 9]^T$. Use CVX to solve this problem with $\alpha = 0.1$, 1, 10, and 100, respectively for $r = 1$, $r = 2$, and $r = 3$, respectively.

(c) Consider the same problem with the difference that the norm is ℓ_3 norm. With the same matrix A and vector b above, solve the problem with CVX for various α values, and plot the optimal value p^* as a function of α, for $r = 1$, 2, and 3.

Solution

Part (a): As an LP,

$$\begin{aligned} \text{minimize:} \quad & \mathbf{1}^T t \\ \text{subject to:} \quad & -t \preceq Ax - b \preceq t \\ & x_{n \text{ largest}} - x_{n \text{ smallest}} \leq \alpha \end{aligned}$$

Part (b):

Problem 5 continued on next page...

r	α	p^*
1	0.1	29.8905
	1	24.3619
	10	0.0000
	100	0.0000
2	0.1	0.0000
	1	0.0000
	10	0.0000
	100	0.0000
3	0.1	0.0000
	1	0.0000
	10	0.0000
	100	0.0000

Part (c):

r	α	p^*
1	0.1	19.5154
	1	16.2028
	10	0.0000
	100	0.0000
2	0.1	0.0000
	1	0.0000
	10	0.0000
	100	0.0000
3	0.1	0.0000
	1	0.0000
	10	0.0000
	100	0.0000

Problem 6

Consider the following problem

$$\begin{aligned} \text{minimize:} \quad & \sum_{i=1}^{k} \phi(|r|_{[i]}) \\ \text{subject to:} \quad & r = Ax - b, \end{aligned}$$

where $|r|_{[1]} \geq |r|_{[2]} \geq \ldots \geq |r|_{[m]}$ are the numbers $|r_1|, |r_2|, \ldots, |r_m|$ sorted in decreasing order. Here matrix A is $m \times n$.

(a) When $\phi(u) = u$, reformulate the problem into an LP. The problem with this penalty function minimizes the sum of the largest k residuals. (*Hint*: you need the formulation in exercise 5.19.)

(b) When $\phi(u)$ is the deadzone-linear penalty function

$$\phi(u) = \begin{cases} 0, & |u| \leq a; \\ |u| - a, & |u| > a; \end{cases}$$

where $a > 0$, reformulate the problem into an LP. This penalty function minimizes the sum of the largest k residuals, but with a deadzone. Recall we have discussed a problem similar to this in class.

Solution

Part (a): As an LP,

$$\begin{aligned} \text{minimize:} \quad & \mathbf{1}^T r \\ \text{subject to:} \quad & r = Ax - b \\ & r \succeq 0 \end{aligned}$$

Part (b):

$$\begin{aligned} \text{minimize:} \quad & \sum_{i=1}^{k} t_i \\ \text{subject to:} \quad & -y \leq |r|_{[i]} \leq y \\ & -t \leq y - |a| \leq t \end{aligned}$$